Ivan Kashuba

Materials Chemist & Process Engineer — Medical Device Materials · Failure Analysis · Process Development

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PROFESSIONAL PROFILE

Materials scientist and chemist-technologist with hands-on laboratory, R&D and production work since 2013 in single crystals, polymers, coatings and monomer processes. My core work is root-cause analysis — working out why a material ended up carrying something it should not: why a batch went out of specification, why a coating delaminated, why a monomer oligomerised, and which change in temperature control, moisture handling or tooling removes the cause rather than the symptom. I prepared, processed and characterised the materials myself, refined the AgBr–AgI phase diagram by differential thermal analysis, and wrote the procedures and process documentation around all of it. I also build the calculation side: engineering calculations rebuilt as reproducible Python with validation and tests, a laboratory-document checker with deterministic decision logic, and day-to-day work with AI coding agents on a live codebase. Co-author of peer-reviewed work on mid-infrared fibre optics. Available for remote contract work.

MEDICAL DEVICE MATERIALS

- **Neonatal phototherapy lamp, in clinical use.** Flexible PMMA light-guide lamp for newborns with hyperbilirubinaemia, developed in a four-person team as contract R&D; a first batch of 100 units went into clinical use. I worked the manufacturing and process side and wrote the assembly workflow.
- **Glycolide, the monomer of bioresorbable polymers.** Thermal-vacuum synthesis and distillation in production, and the root-cause work behind batch quality. Polyglycolide, polymerised from glycolide, is the material of the Neurotube nerve-repair conduit.
- **Coatings on polymer ophthalmic lenses.** Washing, surface activation and adhesion processes, and optical interference coatings; defects investigated against process-log parameters.
- **Flexible surgical light guide.** Single-mode infrared light guides extruded from silver-halide single crystals, on tooling I built; the transmission range was reached, photostability was not.
- **Current focus.** Working through ISO 10993 biological evaluation from the chemistry end: chemical characterisation under part 18, risk reasoning in part 17.

ROOT-CAUSE AND FAILURE ANALYSIS

- **Monomer out of specification — moisture, not the reaction.** Used ^{1}H NMR to identify rapid moisture-associated oligomerisation of glycolide; traced it to ineffective dehydration of raw material and storage-related oxidation rather than to the synthesis step; supported installation of an improved drying cabinet.
- **Oligomer dissolution during filtration — a local hot spot.** Traced unintended dissolution to overheating at the filter element; insulating the element removed it. The practical temperature ranges for thermal-vacuum synthesis and distillation came out of this work.
- **Coating defects on polymer ophthalmic lenses.** Investigated defects with microscopy read against humidity, temperature and equipment parameters from process logs; identified hardened flaking polymer contamination in the finishing area as a defect source and removed it.
- **Powder-coating quality — measured, then corrected.** Judged formulations by gel time, DSC with nitrogen cooling, adhesion and cathodic-disbondment diameter; screened component content, powder humidity and disbondment data for correlation; raised quality by cleaning a liquid heat exchanger, addressing micronisation cleanliness and storage sealing, and improving homogeneity by grinding and sieving.
- **Process window and tooling for hot extrusion.** Studied extrusion modes and tooling strength for two-component silver-halide light guides; designed the tooling in KOMPAS-3D, reviewed ANSYS structural-stress results prepared with a simulation specialist, calculated and built the tubular furnace that heats it.
- **Preparation time cut from weeks to hours.** Reduced preparation of a heat-crosslinking polymer-dye protective solution from about two weeks to a few hours by grinding, controlled heating and mixing.

PROFESSIONAL EXPERIENCE

Independent Research, Engineering Modelling & Scientific Computing — **Feb 2025 – Present**

Independent R&D · Türkiye · Remote

- Convert chemical-engineering calculations from Excel into reproducible Python models with configurable inputs, validation checks, CSV outputs and plots; baselines verified against the source spreadsheets.
- Built and documented an AI laboratory document-automation project combining FastAPI, SQLite, UiPath C# workflows and deterministic PASS / REVIEW / FAIL logic, with human review of uncertain results.
- Develop a multi-module web application using AI coding agents — Claude Code and Codex — for large-codebase navigation, defect location and bounded refactors; write the task specification before the change and verify agent output against tests and repository rules.
- Independent literature research; public GitHub portfolio in materials, polymers, coatings, monomer processes and scientific data workflows.

Research Engineer — Powder Coatings Laboratory Setup & Process Improvement | Institute for Applied Chemistry and Certification FRISHBERG Ltd. — **Jul 2024 – Jan 2025**

Institute for Applied Chemistry and Certification FRISHBERG Ltd. *· Yekaterinburg, Russia*

- Helped establish a powder-coatings R&D laboratory for the corrosion protection of underground metal pipelines: transferred, positioned, connected and commissioned selected production equipment for laboratory use.
- Installed and operated DSC with nitrogen cooling; measured and interpreted thermal transitions and reaction enthalpy in powder-coating systems.

- Investigated replacement components and alternative formulations through gel-time measurement, NMR-solvent selection and laboratory coating trials.
- Processed and visualised data on component content, gel time, powder humidity, cathodic-disbondment diameter and material quantities, with correlation screening.
- Improved temperature control and powder quality: inspected and cleaned a liquid heat exchanger, addressed micronisation cleanliness and storage sealing, raised mixture homogeneity by grinding and sieving.
- Applied tribostatic spraying to heated test specimens; produced an estimated 30 kg of experimental coating batches and assessed adhesion and cathodic disbondment. Turned English equipment manuals into laboratory training procedures.

Chemist-Technologist — Glycolide Monomer Manufacturing & Process Improvement | Medin-N Aug 2021 – Apr 2024
Medin-N · *Yekaterinburg, Russia*

- Maintained technical process documentation and process sheets for glycolide (1,4-dioxane-2,5-dione) synthesis, purification, filtration and thermal-vacuum operations used in bioresorbable and biodegradable polymer production — glycolide is the monomer from which polyglycolic acid (PGA) and PLGA are made.
- Investigated quality deviations related to raw-material dehydration, storage-associated oxidation, filtration and thermal-vacuum processing; identified ineffective dehydration and supported installation of an improved drying cabinet.
- Defined practical temperature ranges for thermal-vacuum synthesis and distillation; improved filtration by insulating the filter element, reducing overheating and unintended oligomer dissolution.
- Contributed to production of an estimated 600 kg of monomer.
- Reduced preparation time for a heat-crosslinking polymer-dye protective solution from about two weeks to a few hours.
- Performed ^{1}H NMR analysis and identified rapid moisture-associated oligomerisation; co-developed a microwave synthesis and vacuum-distillation setup for university project work.

Process Engineer — Optical Coatings | ODV Prescription Optics Plant Apr 2021 – Jul 2021
ODV Prescription Optics Plant · *Yekaterinburg, Russia*

- Supported washing, surface activation and adhesion processes for polymer ophthalmic lenses and optical interference coatings; transferred an interference-coating process to new low-pressure electron-beam deposition equipment.
- Measured polymer moisture content and coating transmission spectra; maintained process logs and investigated coating defects using microscopy together with humidity, temperature and equipment operating parameters.

Research Engineer — Fiber Technology & Photonics | Fiber Technology and Photonics Laboratory, Ural Federal University 2013 – 2021
Fiber Technology and Photonics Laboratory, Ural Federal University · *Yekaterinburg, Russia*

- Alongside my studies, under an employment contract and for part of this period: synthesised AgBr–AgI charge material for crystal growth; researched hot-extrusion parameters for infrared light guides; machined single crystals in a glove box; produced flat infrared lenses by hot pressing on custom tooling; carried out infrared spectroscopy and mechanical testing.
- 2018–2021, in a four-person team, developed a flexible phototherapy lamp for newborns with hyperbilirubinaemia — several bonded layers of PMMA light guides, chosen for their transmission at the required wavelength; the LED source, lenses and control unit already existed, and the flexible lamp was our part of the work. Paid external contract R&D for an industrial manufacturer; a first production batch of 100 units was manufactured and went into clinical use.
- Proposed automated laser engraving of PMMA light guides and identified suitable equipment suppliers; co-assembled an LED UV-curing unit for connector fixation; machined connectors and attachment components.
- Wrote the production and assembly workflow for lamp manufacture.

SCIENTIFIC COMPUTING, ENGINEERING MODELS AND AI-AGENT WORKFLOWS

Public repositories are documented for reproducibility. The calculation projects are educational and theoretical models, not substitutes for validated industrial design calculations.

- **AI Laboratory Document Automation.** Extracts structured values from laboratory documents, validates them against configurable specifications and returns deterministic PASS, REVIEW or FAIL decisions. FastAPI service, SQLite audit history, CSV export, automated tests and UiPath C# workflows for passing, incomplete and out-of-specification scenarios. Human-in-the-loop routing by design: a language model may summarise results but cannot change the rule-engine decision.
- **Reactor dissolution model.** Parameter-driven Python model for agitator-speed scenarios, mass transfer, dissolution time, motor-capacity limits and energy indicators; baseline outputs checked against the source Excel calculation.
- **Pneumatic dryer performance calculator.** Gas velocity, tube diameter, heat transfer, drying time, required pipe length and pressure drop over a configurable operating range, with structured CSV output and visualisation.
- **Stirred-vessel heating comparator.** Compares paddle, anchor and propeller impellers across rotational-speed ranges for heat-transfer performance, heating time and diminishing-return behaviour; baseline checked against the source spreadsheet.
- **Tube-in-tube heat exchanger scaling model.** Converts an Excel-based heating calculation into a parameter-driven Python workflow with criteria-based flow screening, explicit validation and stated limitations.
- **AI coding agents on a live codebase.** Daily use of Claude Code and Codex on a multi-module application: navigating unfamiliar code, locating and debugging defects, bounded refactors, with written specifications before the change and verification against tests. I can describe where agents accelerate technical work, where they produce plausible but wrong results, and how the difference is detected.

EXPERIMENTAL AND PRODUCT DEVELOPMENT PROJECTS

- **Microwave synthesis and vacuum-distillation setup.** Co-developed a laboratory setup for microwave-assisted synthesis and vacuum distillation; selected heating media and process concepts through literature review; used ^{1}H NMR to investigate moisture-associated glycolide oligomerisation.

EDUCATION AND ACADEMIC TRAINING

Postgraduate Programme in Physical Chemistry, completed (dissertation not defended) 2015 – 2019
Ural Federal University, Institute of Chemical Engineering · *GPA 4.4/5.0*

- Diploma qualification awarded: "Researcher. Teacher-Researcher." **No doctoral degree awarded.**
- Research topic: "Research and Development of Photonic-Crystal Single-Mode Light Guides for a Wide Spectral Range."
- Refined the AgBr–AgI phase diagram by differential thermal analysis and investigated compositions for single-mode infrared light guides with a wider transmission range, aimed at a flexible surgical light guide. The transmission range was extended; the material was not photostable and degraded in daylight within days.
- Worked with thermozone crystallisation and Bridgman–Stockbarger growth; machined single crystals in a glove box and hot-extruded them into infrared light guides.
- Designed extrusion tooling in KOMPAS-3D, reviewed ANSYS structural-stress results prepared with a simulation specialist, and calculated and built a tubular furnace for heating the tooling.

Specialist Degree in Chemical Technology — five-year full-time programme 2010 – 2015
Ural Federal University, Institute of Chemical Engineering · *GPA 4.4/5.0*

- Specialisation: chemical technology of single crystals, materials and electronic products.
- Graduation research: "Research and Development of Tooling and Extrusion Modes for Single-Mode and Multimode Infrared Light Guides" — production and hot-extrusion conditions for two-component light guides, crystal machining, tooling strength and optical-quality evaluation.
- Designed and tested extrusion tooling and calculated a tubular furnace integrated with a Shimadzu AGS-X testing machine.

Professional Retraining Diploma — Enterprise Economics and Management 2012 – 2015
Ural Federal University Business School · *Yekaterinburg, Russia*

- Profitability and cost calculations for a small-scale scientific production process, business planning, operations management and industrial decision-making.

Master's Programme — Resource-Saving Technologies and Biotechnology | Ural Federal University 2023 – 2024
Ural Federal University · *first year completed*

- First year completed; discontinued when laboratory work and relocation left no time to continue. Practical coursework in NMR interpretation, chromatographic and spectroscopic methods, molecular docking, protein–ligand interaction analysis, phylogenetic analysis and Excel-based process scale-up equations.

VOLUNTEERING

Volunteer support for children's homes, 2014–2017 — student volunteer activities supporting children in care, alongside study and laboratory work.

PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

Journal articles

- Korsakova E. A., L'vov A. E., **Kashuba I. A.**, Korsakov V. S., Salimgareev D. D., Korsakov A. S., Zhukova L. V. "Fiber-Optic Assemblies Based on Polycrystalline Lightguides for the Mid-IR." *Journal of Optical Technology* 86(7), 439–445 (2019). doi:10.1364/JOT.86.000439 — Russian original in *Оптический журнал*, doi:10.17586/1023-5086-2019-86-07-58-66
- **Kashuba I. A.**, Korsakov A. S., Zhukova L. V., Korsakov V. S. "New Functional Materials for the Infrared Range: Structure, Properties, Application." *Izvestiya Tula State University. Technical Sciences* (2017).

Conference papers and proceedings

- Korsakova E., **Kashuba I.**, Korsakov A., Doroganov S., Skvortsov D., Kalintseva N., Serebryakov V., Zhukova L. "Laser Systems Supplied with Silver Halide Fibres for Laser-Surgery Angioplasty." *Proc. 2019 USBEREIT*, IEEE. doi:10.1109/USBEREIT.2019.8736633
- Lashova A. A., Korsakov V. S., Korsakova E. A., **Kashuba I. A.**, Zhukova L. V. "Spectral Oil Analyzer." *Journal of Physics: Conference Series* 1421(1), 012056 (2019). doi:10.1088/1742-6596/1421/1/012056
- Korsakova E., Muftahitdinova N., Korsakov V., **Kashuba I.**, Korsakov A., Zhukova L. "Infrared Fiber Bundles Based on Silver Halide Crystals for Monitoring and Control Systems in Heat and Power Engineering." *DEStech Transactions on Environment, Energy and Earth Sciences* (2019). doi:10.12783/dteees/tpcase2018/30406
- Korsakova E. A., Muftahitdinova N. A., **Kashuba I. A.**, Korsakov A. S., Zhukova L. V. "Silver Halide IR Fibre with Extra Small Diameter." *Proceedings of ITNT-2019*, pp. 566–572.
- Salimgareev D. D., L'vov A. E., Korsakov V. S., Gul'ko D. Y., **Kashuba I. A.**, Korsakov A. S., Zhukova L. V. "β-Radiation-Resistant Optical Fibers Based on Modified Silver Halide Crystals." *17th International Conference-School on Materials of Nano-, Micro-, Optoelectronics and Fiber Optics*, Saransk, 2018, p. 40. Conference abstract.

TECHNICAL METHODS AND EQUIPMENT

Materials and processes	Silver-halide single crystals; thermozone crystallisation; Bridgman–Stockbarger growth; glove-box machining; hot extrusion and hot pressing; powder-coating formulation and tribostatic application; glycolide synthesis; thermal-vacuum processing and distillation; filtration; drying; micronisation; electron-beam vacuum deposition; optical-surface cleaning, activation and adhesion.
Analytical methods	Hands-on use and interpretation: ^{1}H NMR, DSC (with nitrogen cooling), DTA, TGA, FTIR, UV-Vis spectrophotometry, microscopy, moisture, pH, conductivity and thickness measurement. Introductory training: ^{13}C NMR, HSQC, HMBC, TLC, HPLC, LC-MS, GC-MS, UPLC.
Equipment	DSC with nitrogen cooling; thermal-vacuum reactors and vacuum systems; electron-beam deposition equipment; powder-coating and micronisation line; extrusion tooling and Shimadzu AGS-X testing machine; CO_2 laser; UV-curing equipment; microwave reactor; glove box; ultrasonic bath; centrifuge; laboratory drying, grinding and heating equipment.

Quality and test work	Gel-time measurement; cathodic-disbondment evaluation; adhesion and coating trials; spectral-transmission measurement; experimental batch preparation; process-deviation investigation; procedure development and technical documentation.

SOFTWARE AND LANGUAGES

Established use	Microsoft Excel; Origin; MestReNova; ChemDraw; Chem3D; KOMPAS-3D; Microsoft Office and Google Workspace; scientific literature search; technical documentation and presentation development.
Developing use	Python (engineering calculations, validation, plotting), Google Colab, Git and GitHub, JSON/CSV workflows, UiPath Studio, FastAPI, SQLite, AI coding agents (Claude Code, Codex) for coding, research and documentation workflows.
Databases and modelling	Reaxys; Google Scholar; PubMed and NCBI; RCSB PDB; ProteinsPlus; PyMOL; AutoDockTools; Open Babel; SwissTargetPrediction. ANSYS: review and interpretation of structural-analysis results, not independent expert-level simulation.
Languages	Russian — native · English — B2, professional working proficiency